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SRM

DEPARTMENT OF CHEMISTRY
College of Engineering and Technology
SRM Institute of Science and Technology
Kattankulathur – 603203

Set-2

INTERNAL ASSESSMENT – II [CLA1 T2]

Program: B.Tech	**Date: 13/10/2023**
Course Code & Title:21CYB101J & Chemistry	**Time: 8.00-9.00am**
Year &Sem: I Year & I Sem	**Max. Marks: 30 marks**

Part – A (10 x 1 = 10 Marks)

Answer ALL the Questions

1. Ability of atom to attract electrons towards itself is called
   (a) Electron affinity
   (b) Electronegativity
   (c) Ionization energy
   (d) Shielding effect
2. IE_1 and IE_2 of Mg are 179 and 348 Kcal mol^{-1} respectively. The energy required for the reaction $Mg \rightarrow Mg^{2+} + 2e^-$ is
   (a) + 169 Kcal mol^{-1}
   (b) - 169 Kcal mol^{-1}
   (c) + 527 Kcal mol^{-1}
   (d) - 527 Kcal mol^{-1}
3. According to Pearson's HSAB concept, the product is more stable when,
   (a) Hard acid combines with soft base.
   (b) Soft acid combines with Hard base
   (c) Hard acid combines with Hard base
   (d) cannot be predicted
4. The HSAB theory can be used to predict solubility trends. Based on the determination of the hardness of the ions, what happens when AgF and LiI are placed together into solution
   (a) Only AgI precipitates initially, but dissolves back into solution when Ag+ forms a complex ion with F− and H2O.
   (b) Only LiF precipitates
   (c) Both LiF and AgI precipitate
   (d) No precipitate is formed
5. The correct relationship between free energy change in a reaction and the corresponding equilibrium constant K_C is
   (a) $-\Delta G = RT \ln K_C$
   (b) $\Delta G^0 = RT \ln K_C$
   (c) $-\Delta G^0 = RT \ln K_C$
   (d) $\Delta G = RT \ln K_C$
6. All the naturally occurring processes proceed spontaneously in a direction that leads to
   (a) decrease of entropy
   (b) increase of enthalpy
   (c) an increase of free energy

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(d) decrease of free

7. If $E°_{Fe2+/Fe} = -0.441$ V and $E°_{Fe3+/Fe2+} = 0.771$ V, the standard EMF of the reaction, $Fe + 2Fe^{3+} \rightarrow 3Fe^{2+}$ will be
   (a) 1.212 V
   (b) 0.111 V
   (c) 0.330 V
   (d) 1.653 V
8. When equilibrium is reached inside the two half-cells of the electrochemical cells, what is the net voltage across the electrodes?
   (a) > 1
   (b) < 1
   (c) = 0
   (d) Not defined
9. Leakage of inflammable gas from a corroded pipe can cause _____
   (a) Acidity
   (b) Alkalinity
   (c) Turbidity
   (d) Fire hazards
10. The solubility product increases with an increase in
   (a) Energy
   (b) Temperature
   (c) Pressure
   (d) Volume

Part – B (2 x 10 = 20 Marks)

11. (a) i. Explain with examples the variation of any three periodic properties along the period and down the group? (6 Marks)
   ii. Write a note on applications and limitations of HSAB principle. (4 Marks)

(OR)

(b) i. Derive Gibbs-Helmholtz equation. (6 Marks)
   ii. Arrange the compounds in order of increasing entropy (S) and give reasons. (4 Marks)
   $CH_3OH(s)$, $CH_3OH(l)$, $CH_3OH(g)$, and $CH_3CH_2OH(g)$

12. (a) i. Construct an Electrochemical cell and give its representations. (4 Marks)
   ii. The molar solubility of $PbBr_2$ is 2.17×10^{-3} mol/L at a certain temperature. Calculate K_{sp} for $PbBr_2$ (6 Marks)

(OR)

(b) With a neat sketch, explain Pourbaix diagram for Iron system. (10 Marks)

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Set-1

INTERNAL ASSESSMENT – II [CLA1 T2]

Program: B.Tech
Course Code & Title:21CYB101J & Chemistry
Year &Sem: I Year & I Sem

Date: 13/10/2023
Time: 8.00-9.00AM
Max. Marks: 30 marks

Part – A (10 x 1 = 10 Marks)

Answer ALL the Questions

1. Atomic radii _____ along the periods
   (a) Increases
   (b) Decreases
   (c) Remains constant
   (d) Irregular
2. On the Pauling's electronegativity scale the element next to F is
   (a) N
   (b) Cl
   (c) O
   (d) Ne
3. HSAB concept can be used to determine,
   (a) Stability of complexes
   (b) Predicting feasibility of reactions
   (c) Solubility of compounds in a given solvent
   (d) All the above
4. Which of the following is not a hard acid ?
   (a) H^+
   (b) Li^+
   (c) BH_3
   (d) BF_3
5. Which of the following is not a thermodynamic function?
   (a) Internal energy
   (b) Enthalpy
   (c) Entropy
   (d) Frictional energy
6. The amount of heat exchanged with the surrounding at constant temperature and pressure is given by the quantity
   (a) ΔE
   (b) ΔH
   (c) -I
   (d) ΔG
7. When heating one end of a metal plate, the other end gets hot because of
   (a) The resistance of the metal
   (b) Mobility of atoms in the metal
   (c) Energised electrons moving to the other end
   (d) minor perturbation in the energy of atoms.
8. $Ag^+(aq) + e^- \rightarrow Ag(s)$ $E^\circ = +0.80$ V

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$Fe^{2+}(aq) + + 2e^- \rightarrow Fe(s)$ $E^\circ = -0.44$ V

What is the emf of the given reaction?

$Fe(s) + 2Ag^+(aq) \rightarrow Fe^{2+}(aq) + 2Ag(s)$

(a) 1.16 V
(b) 1.24 V
(c) 2.04 V
(d) -1.16 V

9. Which of the following metal is resistant to corrosion?
   (a) Titanium
   (b) Zinc
   (c) Magnesium
   (d) Iron

10. In electrochemical corrosion
   (a) Anode undergoes oxidation
   (b) Cathode undergoes oxidation
   (c) Both undergo oxidation
   (d) None undergoes oxidation

Part – B (2 x 10 = 20 Marks)

11. (a) i. What is electron affinity? What are the factors influencing it and arrange Br, F, I and Cl in the order of increasing electron affinity. (4 Marks)

   ii. According to Pearson's rule distinguish between (6 Marks)
   A) Hard acid and soft acid B) Hard base and soft base

(OR)

   (b) i. Calculate the Gibbs free energy for a reaction which involves the change in enthalpy as 19.07 Kcal and change in entropy as 90 cal/K at 293K (4 Marks)

   ii Derive Nernst equation and give its applications. (6 Marks)

12. (a) i. Give an account of Hydrogen evolution type of wet corrosion (4 Marks)

   ii. What is the expression for the solubility product constant of common salt? What factors affect the value of Ksp? (6 Marks)

(OR)

   (b) i. What is Internal energy? Give its relation with enthalpy.

   ii. What is the Cell Potential of the electrochemical cell for the given cell reaction?
   $Pb^{2+} + Cd \rightarrow Pb + Cd^{2+}$
   Given that E^0 cell = 0.277 volts, Temperature = 25°C, $[Cd^{2+}] = 0.02$ M, and $[Pb^{2+}] = 0.2$ M

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College of Engineering and Technology
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Kattankulathur – 603203

Set-2

RA2311009010078

INTERNAL ASSESSMENT – II [CLA1-T2]

Program: B.Tech
Course Code & Title:21CYB101J & Chemistry
Year &Sem: I Year & I Sem

Date: 17/10/ 2023
Time: 12.30-1.30PM
Max. Marks: 30 marks

Part – A (10 x 1 = 10 Marks)

Answer ALL The Questions

1. BH_3 gives a stable complex with -----.
   a) CO b) Pt^{2+} c) OH^- d) methane
2. One of the following is an example for hard acid?
   a) Cs^+ b) Li^+ c) Chloride d) BH_3
3. The correct order for the size of I, I^+, I^- is---- [Z for I =53]
   a) $I > I^- > I^+$ b) $I > I^+ > I^-$ c) $I^- > I > I^+$ d) $I^+ > I^- > I$
4. Which one is the smallest ion?
   a) Al^{3+} b) Mg^{2+} c) Na^+ d) Ba^{2+}
5. Heat does not spontaneously flow from a colder body to a hotter one. Which of the following thermodynamics law states this?
   a) Zeroth law of thermodynamics b) first law of thermodynamics c)second law of thermodynamics d)third law of thermodynamics
6. For a reaction at 127 ^{0}C, the ΔH and ΔS are 92.22 kJ/mol and -198.75 J/K.mol respectively. Then it ------.
   a) is spontaneous b) is not spontaneous c) may be spontaneous d) is non predictable.
7. Tarnishing of Ag ware in H_2S laden air is an example of ---- corrosion.
   a) Dry b) oxygen absorption type c) galvanic d) hydrogen evolution type
8. Find the number of electrons transferred in the equation:
   $Cu(g) + 2Ag^+(aq) \rightarrow Cu^{2+}(aq) + 2Ag(s)$.
   a) 4 b)3 c)2 d)1
9. Using the data given below, find out the most stable ion in its reduced form.
   $E^0\ Cr_2O_7^{2-}/Cr^{3+} = 1.33V$
   $E^0\ MnO_4^-/Mn^{2+} = 1.51V$
   $E^0\ Cl_2/Cl^- = 1.36V$
   $E^0\ Cr^{3+}/Cr = -0.74V$
   a) Cl^- b) Cr^{3+} c) Cr d) Mn^{2+}
10. Which of the following conditions are satisfied when the cell reaction in the electrochemical cell is spontaneous?
    a) $\Delta G^\circ > 0$ b) $E^\circ cell < 0$ c) $E^\circ cell = 0$ d) $\Delta G^\circ < 0$

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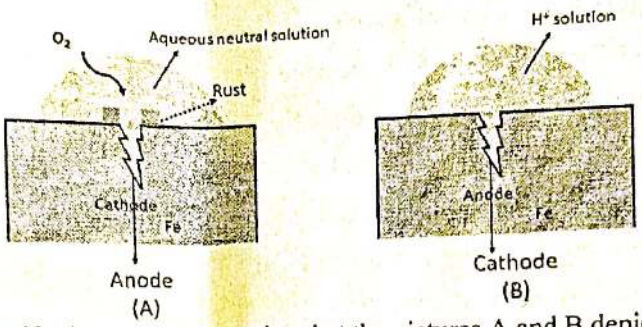
Part – B (2 x 10 = 20 Marks)

11. a.i. Compare hard and soft acids with suitable examples. (4 Marks)

ii. The electron affinity of F is -328 $kJmol^{-1}$ which is higher than that of Oxygen -141 $kJ\ mol^{-1}$, but Chlorine has -349 $kJmol^{-1}$ which is higher than Fluorine. Give reasons for these discrepancies. (6 Marks)

(OR)

b.



I. Identify the types of corrosion that the pictures A and B depicts? (1 Mark)

II. Give the anodic and cathodic reactions for A and B type of corrosions. (4 Marks)

III. What happens if oxygen supply is increased in "A" type of corrosion? (2 Marks)

IV. Why rust rests on cathode not inside anode in "A" type of corrosion? (2 Marks)

V. From observing "B" type of corrosion, which metals in EMF series readily displaces hydrogen ions? (1 Mark)

12. a. i With the help of Pourbaix diagram for Iron system, explain potential dependent reactions. (6 Marks)

ii. Give the solubility expression and calculate the solubility of Magnesium hydroxide. [Solubility product constant (K_{sp}) = 2.00 x 10^{-11}]. (4 Marks)

(OR)

b. Derive Nernst equation and give its applications.

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INTERNAL ASSESSMENT – II

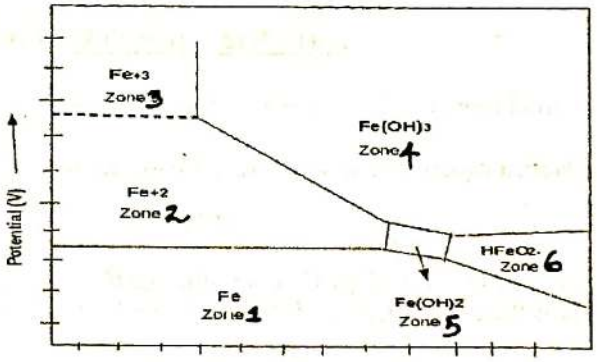
Set-1

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & II Sem

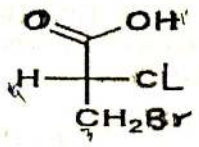
Date: 27-03-2023
Time: 12.30 – 1.30 PM
Max. Marks: 30 marks

Part – A (10 x 1 = 10Marks)
Answer ALL The Questions

1. All the naturally occurring processes proceed spontaneously in a direction that leads to
   a. decrease of entropy b. increase of enthalpy c. an increase of free energy d. decrease of free energy
2. If spontaneity of a reaction is temperature dependent and endothermic, then
   a. $\Delta H<0$ and $\Delta S>0$ b. $\Delta H<0$ and $\Delta S<0$ c. $\Delta H>0$ and $\Delta S>0$ d. $\Delta H>0$ and $\Delta S<0$.
3. The e.m.f and the standard e.m.f of a cell in the following reaction is 5 V and 5.06 V at room temperature, $Ni(s) + 2Ag^{+}_{(n)} \rightarrow Ni^{2+}(0.02M) + 2Ag_{(s)}$. What is the concentration of Ag^{+} ions?
   a) 0.0125 M b) 0.0314 M c) 0.0625 M d) 0.0174 M
4. Metals do not exist in nature in the form of-----
   a) Nitrates b) Sulphates c) Carbonates d) Oxides
5. Which of the following zone of the given Pourbaix diagram indicates the passive zone?



   a) Zone 1 and 2 b) Zone 2 and 3 c) Zone 2, 3 and 6 d) Zone 4 and 5
6. Assign R/S notation for the given compound:



   a) R b) S c) Neither R nor S d) R or S

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7. Diastereomers are

   a) Geometrical isomers b) Mirror images c) Non-mirror images d) Unstable molecules

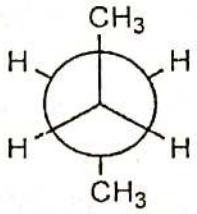
8. The water molecule exhibits .....rotation axis of symmetry.

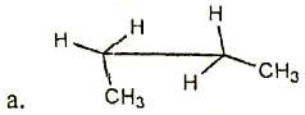
   a) C_3 b) C_2 c) C_4 d) C_n

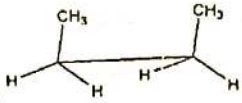
9. The potential energy of n-butane is maximum for _____

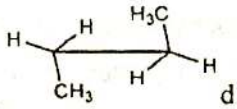
   a) Skew conformations b) Staggered conformations c) Eclipsed conformations d) Gauche

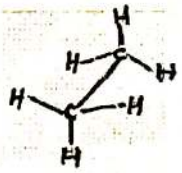
10. Which of the following sawhorse representations is correct for the given Newman projection?



a. 

b. 

c. 

d. 

Part – B (2 x 10 = 20 Marks)

11. a. i. Define internal energy and enthalpy of a system. Give their relation. (4 Marks)

    ii. How the Nernst equation is applied to acid-base and redox reactions. (6 Marks)

(OR)

b. i. Define solubility product (Ksp). Calcium fluoride (CaF_2) dissolves in water to the extent of 0.00170 g/100ml. What is the Ksp for CaF_2? [MM for CaF_2=78.1 g/mol] (4 Marks)

   ii. Explain the mechanism involved in oxidation corrosion. (6 Marks)

12. a. i. Give the differences between mesomers and racemic mixture. (4Marks)

    ii. Brief on the projections and sketch it for the following compounds. (6 Marks)
    A. Staggered form of Sawhorse projection for 2-butanol.
    B. Fisher projection for L-form of lactic acid.
    C. Staggered form of Newman projection for CH_3-CH_3.

(OR)

b. Explicate structural isomerism exhibited by organic com pounds with suitable examples.

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SET 1

DEPARTMENT OF CHEMISTRY
SRM Institute of Science and Technology
College of Engineering and Technology
Kattankulathur – 603203
Internal Assessment – II

Program: B. Tech | **Date: 29-03-23**
Course Code & Title: 21CYB101J & Chemistry | **Time: 8.00 – 9.00 AM**
Year & Sem: I Year & II Sem | **Max. Marks: 30 marks**

Part – A (10 x 1 = 10Marks)
Answer ALL The Questions

1. The equilibrium constant expression (K_{sp}) for $Ba_3(PO_4)_{2(s)}$ is -----

   a) $[Ba^{2+}]^3.[PO_4^{3-}]^2$ b) $3[Ba^{2+}].[PO_4^{3-}]$ c) $[Ba^{2+}].[PO_4^{3-}]$ d) $[Ba^{2+}].3[PO_4^{3-}]$

2. Which of the following is NOT correct?

   a) ΔG is zero for a reversible reaction. b) ΔG is positive for a spontaneous reaction.

   b) ΔG is negative for a spontaneous reaction. c) ΔG is positive for a non-spontaneous reaction.

3. What is the entropy change (in $JK^{-1}\ mol^{-1}$) when 1 mole of ice is converted into water at 0^0C?
   (The enthalpy change for the conversion of ice to liquid water is 6.0 kJ mol^{-1} at 0^0C)

   a) 20.13 b) 2.013 c) 2.198 d) 21.98

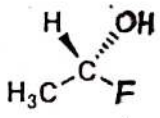
4. Passivity is due to

   a) Higher EMF b) Lower EMF c) Oxide film d) both a and b

5. Stable layer is formed by one of the following oxides:

   a) MoO_3 b)Al_2O_3 c)Na_2O d) CaO

6. Assign R/S notation for the given compound:



   a) R b) S c) Neither R nor S d) R or S

7. Which one of the following conformations of cyclohexane is chiral?

   a) Boat b)Twist Boat c)Chair d) None of the above

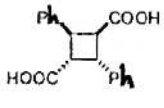
8. Which of the following alkane that can exhibit optical activity?

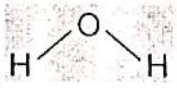
   a) Neopentane b)Isopentane c)3–Methylpentane d)3–Methylhexane

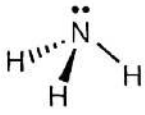
9. The Gauche conformation(s) for n-butane occurs at angle(s) -------.

   a) 60^0 and 300^0 b) 360^0 c) 180^0 and 120^0 d) only at 300^0

10. The following compound that has centre of symmetry is-----.

a)  b) 

c) 

d) 

Part-B (2x 10= 20 Marks)

11.a. i. Compare hydrogen evolution and oxygen absorption type of wet corrosion. (6 Marks)

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ii. Calculate the enthalpy of formation of carbon di sulphide; given that the enthalpy of combustion of it is 110.2 kJ mol^{-1} and those of Sulfur and Carbon are 297.4 kJ and 394.5 kJ/g atoms, respectively. (4 Marks)

(OR)

b. With a neat sketch of Pourbaix diagram for Iron system, explain redox reactions and acid-base reactions.

12. i. Expound CIP priority rules to assign R/S notation for a molecule having one or two chiral centre(s).

OR

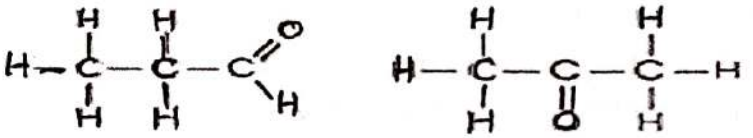
ii. Identify the isomerism exhibited by the pairs given below and write a note on the isomerism.

A. $CH_3CH_2—O—CH_2CH_3$ $CH_3—O—CH_2CH_2CH_3$

B.



C.



D.



E.



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SET 1

SRM Institute of Science and Technology
College of Engineering and Technology
Department Of Chemistry
SRM Nagar, Kattankulathur – 603203

ANSWER KEY

Internal Assessment – II

Program: B.Tech | **Date: 27.03.2023**
Course Code & Title: 21CYB101J/Chemistry | **Time: 08.00-9.00 am**
Year & Sem: I/II | **Max. Marks: 30 Marks**

Part-A (10x1=10Marks)

1. Gibbs function G is given by
   a) H-TS

2. In a reversible process, entropy of the system
   d) remains constant

3. What is meant by corrosion?
   c) Both a. and b.

4. The process of transmitting electric current through an electrolyte solution it is known as
   c) Electrolysis

5. The name of the equation showing relation between electrode potential standard potential (Eo) and concentration of ions in solution is
   b) Nernst equation

6. If our eyes travel in clockwise direction from the ligand of highest priority to the ligand of lowest priority, the configuration is
   a) R-Configuration

7. Identify the chiral molecule among the following
   d) Isobutyl alcohol

8. 2-Chloro propane and 1-Chloro propane exhibit…..Isomerism
   b) Position

9. The potential energy of n-butane is minimum for
   b) Staggered conformation

10. Chiral molecules which are non- super-imposable mirror images of each other are called
    d) Enantiomers

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Part-B (2x 10= 20 Marks)

11. (a) i Define the terms (6Marks)

1. Entropy-Definition [3 x 2=6Marks]

Ans: Entropy is defined as a measure of randomness or disorder of a system

2. Enthalpy-Definition

Ans: Enthalpy is the measurement of energy in a thermodynamic system. The quantity of enthalpy equals to the total content of heat of a system, equivalent to the system's internal energy plus the product of volume and pressure

3. Internal energy-Definition

Ans: *An energy form inherent in every system is the internal energy*, which arises from the molecular state of motion of matter. The symbol U is used for the internal energy and the unit of measurement is the joules (J).

(ii) Calculate ΔG for the following reaction at 25 °C. Will the reaction occur spontaneously? (4Marks)

NH_3 (g) + HCl (g) → NH_4Cl(s)

ΔH = -176.0KJ

ΔS = -284J/K

Ans: Convert Δ S = KJ/K (OR) ΔH into J, Temperature into Kelvin

The definition of Gibbs energy can then be used directly

$\Delta G=\Delta H-T\Delta S$

$\Delta G=-176.0kJ-(298K)(-0.284.8kJ/K)$

$\Delta G=-176.0kJ-(-84.9kJ)$

ΔG=−91.1kJYes, this reaction is spontaneous at room temperature since ΔG is negative.

(OR)

(b) With a neat sketch explain Pourbaix diagram for Iron.

Explanation with diagram and equations. [8+2=10 marks]

Some important points are highlighted:

- Pourbaix Diagrams plot electrochemical stability for different redox states of an element as a function of pH. As noted above, these diagrams are essentially phase diagrams that plot the map the conditions of potential and pH (most typically in aqueous solutions) where different redox species are stable.
- Areas in the Pourbaix diagram mark regions where a single species (Fe^{2+}(aq), Fe_3O_4(s), etc.) is stable. More stable species tend to occupy larger areas.
- Lines mark places where two species exist in equilibrium.
- Pure redox reactions are horizontal lines - these reactions are not pH-dependent
- Pure acid-base reactions are vertical lines - these do not depend on potential
- Reactions that are both acid-base and redox have a slope of -0.0592 V/pH x (nH^+/me^-)

Examples of equilibria in the iron Pourbaix diagram (numbered on the plot):

$Fe^{2+}+2e^-\rightarrow Fe_{(s)}$ (pure redox reaction - no pH dependence) - No 1

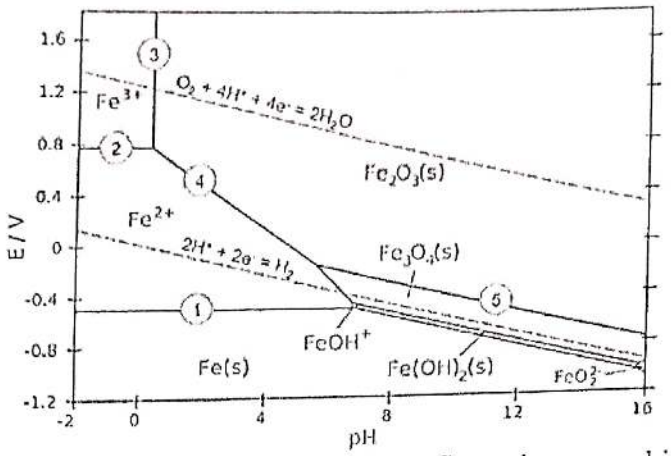
$Fe^{3+}+e^-\rightarrow Fe^{2+}$ (pure redox reaction - no pH dependence) - No 2

$2Fe^{3+}+3H_2O\rightarrow Fe_2O_3(s)+6H^+$ (pure acid-base, no redox) - No 3

$2Fe^{2+}+3H_2O\rightarrow Fe_2O_3(s)+6H^+ + 2e^-$ (slope = -59.2 x 6/2 = -178 mV/pH) - No 4

$2Fe_3O_4(s)+H2O\rightarrow 3Fe_2O_3(s)+2H^++2e^-$ (slope = -59.2 x 2/2 = -59.2 mV/pH) - No 5

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12. (a), Explain CIP priority rules to determine R/S Configuration on a chiral center taking an example [8+2=10 marks]

Rule 1

First, examine at the atoms directly attached to the stereocenter of the compound. A substituent with a higher atomic number takes precedence over a substituent with a lower atomic number. Hydrogen is the lowest possible priority substituent, because it has the lowest atomic number.

1. When dealing with isotopes, the atom with the higher atomic mass receives higher priority.

2. When visualizing the molecule, the lowest priority substituent should always point away from the viewer (a dashed line indicates this). To understand how this works or looks, imagine that a clock and a pole. Attach the pole to the back of the clock, so that when when looking at the face of the clock the pole points away from the viewer in the same way the lowest priority substituent should point away.

3. Then, draw an arrow from the highest priority atom to the 2nd highest priority atom to the 3rd highest priority atom. Because the 4th highest priority atom is placed in the back, the arrow should appear like it is going across the face of a clock. If it is going clockwise, then it is an R-enantiomer; If it is going counterclockwise, it is an S-enantiomer.

Rule 2

If there are two substituents with equal rank, proceed along the two substituent chains until there is a point of difference. First, determine which of the chains has the first connection to an atom with the highest priority (the highest atomic number). That chain has the higher priority.

If the chains are similar, proceed down the chain, until a point of difference.

Rule 3

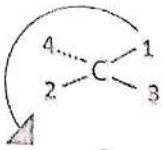
If a chain is connected to the same kind of atom twice or three times, check to see if the atom it is connected to has a greater atomic number than any of the atoms that the competing chain is connected to.

- If none of the atoms connected to the competing chain(s) at the same point has a greater atomic number: the chain bonded to the same atom multiple times has the greater priority

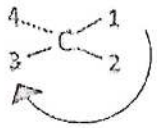
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- If however, one of the atoms connected to the competing chain has a higher atomic number: that chain has the higher priority.

Stereocenters are labeled R or S The "right hand" and "left hand" nomenclature is used to name the enantiomers of a chiral compound. The stereocenters are labeled as R or S.



S



R

(OR)

(b). i. Write a notes on Optical isomerism [5Marks]

Optical isomerism is a type of stereoisomerism in which the isomers have the same molecular formula and the structural formula but differ in their direction of rotation of plane polarized light.

- A solution of one enantiomer rotates the plane of polarisation in a clockwise direction. This enantiomer is known as the (+) form.
- A solution of the other enantiomer rotates the plane of polarisation in an anti-clockwise direction. This enantiomer is known as the (-) form.
- If the solutions are equally concentrated the amount of rotation caused by the two isomers is exactly the same - but in opposite directions.

ii. Differentiate between Enantiomers and Diasteromers. (5Marks)

Ans:

Enantiomers	Diastereomers
Enantiomers are stereoisomers that are mirror images of each other	Diastereomers are stereoisomers that aren't a mirror image of each other
Have identical physical properties except for the ability to rotate plane-polarised light	Distinct physical properties
Present in pairs	There can be several molecules
Similar molecular shape	Different molecular shape

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Set-1

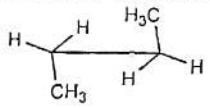
INTERNAL ASSESSMENT – II
ANSWER KEY

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & II Sem

Date: 27.03.2023
Time: 12.30 – 1.30 PM
Max. Marks: 30 marks

Part – A (10 x 1 = 10Marks)

1. d. decrease of free energy
2. c. $\Delta H>0$ and $\Delta S>0$
3. d) 0.0174 M
4. a) Nitrates
5. d) Zone 4 and 5
6. b) S
7. c) Non-mirror images
8. b) C_2
9. c) Eclipsed conformations
10. c.



Part – B (2 x 10 = 20 Marks)

11. a. i. Define internal energy and enthalpy of a system. Give their relation. (4 Marks)

The sum of all the energies contained in a system, such as mechanical, chemical, potential, kinetic, and so on, is its internal energy. Internal energy is denoted by the letter U, and joules are the unit of measurement (J).
Enthalpy is a measure of the total heat energy, absorbed or released, in a thermodynamic system. [2]
Relation between U and H. [2]

ii. How the Nernst equation is applied to acid-base and redox reactions. (6 Marks)

Application towards acid-base and redox reactions using Nernst equation. (3x2=6 Marks)

(OR)

b. i. Define solubility product (Ksp). Calcium fluoride (CaF_2) dissolves in water to the extent of 0.00170 g/100ml. What is the Ksp for CaF_2? [MM for CaF_2=78.1 g/mol] (4 Marks)

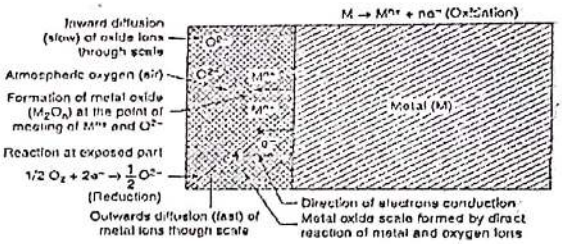
Definition. (1 Mark)

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Solubility of $CaF_2 = 1.7 \times 10^{-3} g/100mL = 17 \times 10^{-3} gL^{-1}$
Molar mass of $CaF_2 = (40 + 2 \times 19) = 78 gmol^{-1}$
Solubility of $CaF_2 = \dfrac{(17 \times 10^{-3} gL^{-1})}{(78.0 gmol^{-1})} = 2.18 \times 10^{-4} molL^{-1}$
The solubility equibrium for the saturated solution of CaF_2 is :
$CaF_2(s) \overset{aq}{\Leftrightarrow} Ca^{2+}(aq) + 2F^-(aq)$
$[Ca^{2+}(aq)] = 2.18 \times 10^{-4} molL^{-1}$
$[F^-(aq)] = 2 \times 2.18 \times 10^{-4} molL^{-1}$
Applying Law of chemical equilibrium :
$K_{sp} = [Ca^{2+}(aq)][F^-(aq)]^2$
$= (2.18 \times 10^{-4} molL^{-1}) \times (2.18 \times 10^{-4} molL^{-1})^2 = 4.14 \times 10^{-11} mol^3.$

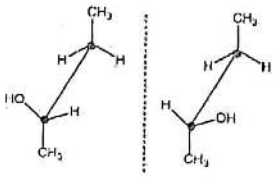
(3 Marks)

ii. Explain the mechanism involved in oxidation corrosion. (6 Marks)
Mechanism with equations and explanation. **(5)**
Diagram. **(1)**

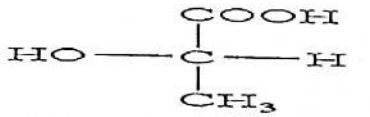


12. a. i. Give the differences between mesomers and racemic mixture. (4Marks)
`Any 2 differences with examples. **(2x2=4Mks)**

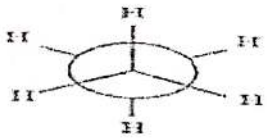
ii. Brief on the projections and sketch it for the following compounds. (6 Marks)
Brief on each projection. **(3 Marks)**
Projections **(3 Marks)**



A. 2-butanol



B. L(+) lactic acid



C. CH_3-CH_3

(OR)

b. Explicate structural isomerism exhibited by organic compounds with suitable examples.
Explanation on all five isomerism with examples. **(5x2=10 Marks)**

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Set-2

INTERNAL ASSESSMENT – II

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & II Sem

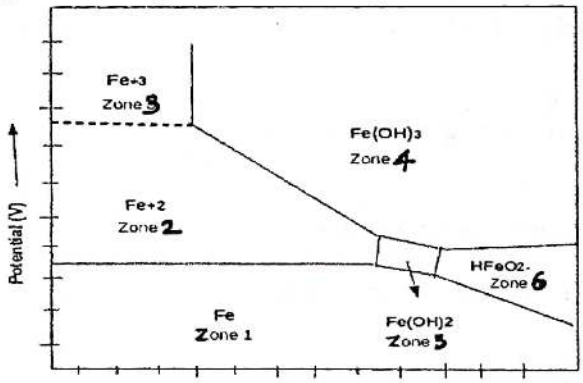
Date: 27-03-2023
Time: 12.30 – 1.30 PM
Max. Marks: 30 marks

Part – A (10 x 1 = 10Marks)
Answer ALL The Questions

1. Which of the following comes under wet corrosion?

a) Concentration cell corrosion b) Oxidation corrosion c) Liquid metal corrosion d) Corrosion by other gases

2. Which of the following zone in the given Pourbaix diagram indicates the immune zone?



a) Zone 1 b) Zone 2 c) Zone 3 d) Zone 4

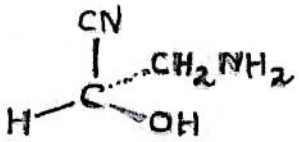
3. Calculate the Gibbs free energy for the reaction of conversion of ATP into ADP at 293 Kelvin the change in enthalpy is 19.07 kilocal and the change in entropy is 90 cal per Kelvin.
   a) 7.3 Cal b) –5.3Kcal c) 7.3 Kcal d) –7.3Kcal
4. For a spontaneous process, the change in Gibbs function is equal to .....
   a) the heat content of the system b) entropy change of the system c) work of expansion d) useful work
5. MoO_3 forms ..... type of layer in a dry atmosphere.
   a) stable b) unstable c) volatile d) porous
6. Which of the following is NOT an example of a chiral object?

a)  b)  c)  d) 

7. When the compound n-butane is rotated to an angle of 240^0 then it forms------ conformation.
   a. staggered b. fully eclipsed c. eclipsed form B d. Gauche

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8. A center of symmetry is equivalent to _____ fold alternating axis of symmetry.
   a) One b) Two c) Three d) Four
9. Assign R/S notation for the following compound.



   a. R b. S c. Neither R nor S d. R or S
10. If our eyes travel in a counter-clockwise direction from the ligand of highest priority to the ligand of lowest priority, the configuration is
    a) R-Configuration b) S-Configuration c) E-Configuration d) Z-Configuration

Part – B (2 x 10 = 20 Marks)

11. a. i. Define Gibbs free energy and Helmholtz work function. Prove $\Delta G = \Delta A + P\Delta V$. (6 Marks)

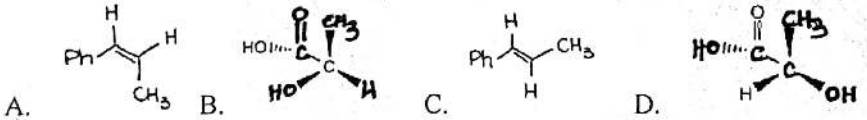
    ii. For the reaction $Ag_2O_{(s)} \longrightarrow 2Ag_{(s)} + 0.5O_{2\,(g)}$, $\Delta H = 30.56$ KJ/mol; $\Delta S = 6.66$ J/K/mol. Calculate the temperature at which $\Delta G = 0$? (4 Marks)

    (OR)

    b. i. Explain the Pilling-Bedworth rule. (4 Marks)

    ii. Describe the oxygen absorption type of electrochemical corrosion with a suitable mechanism. (6 Marks)

12. a. i. Explain the terms enantiomers and diastereomers. Which of the following the pair of compounds shown, best describes enantiomers, diastereomers? (6 Marks)



    ii. Write notes on Geometrical isomerism. (4 Marks)

    (OR)

    b. Explain the following with suitable examples.
    i. Plane of Symmetry ii. Axis of Symmetry iii. Tautomerism iv. Chain isomerism and v. Inversion Symmetry

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Set-1

INTERNAL ASSESSMENT – II [CLA1 T2]

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year &Sem: I Year & I Sem

Date: 13/10/2023
Time: 8.00-9.00AM
Max. Marks: 30 marks

Answer ALL the Questions

Part – A (10 x 1 = 10 Marks)

1. Atomic radii _____ along the periods
   (a) Increases
   (b) Decreases
   (c) Remains constant
   (d) Irregular
2. On the Pauling's electronegativity scale the element next to F is
   (a) N
   (b) Cl
   (c) O
   (d) Ne
3. HSAB concept can be used to determine,
   (a) Stability of complexes
   (b) Predicting feasibility of reactions
   (c) Solubility of compounds in a given solvent
   (d) All the above
4. Which of the following is not a hard acid ?
   (a) H^+
   (b) Li^+
   (c) BH_3
   (d) BF_3
5. Which of the following is not a thermodynamic function?
   (a) Internal energy
   (b) Enthalpy
   (c) Entropy
   (d) Frictional energy
6. The amount of heat exchanged with the surrounding at constant temperature and pressure is given by the quantity
   (a) ΔE
   (b) ΔH
   (c) -I
   (d) ΔG
7. When heating one end of a metal plate, the other end gets hot because of
   (a) The resistance of the metal
   (b) Mobility of atoms in the metal
   (c) Energised electrons moving to the other end
   (d) minor perturbation in the energy of atoms.
8. $Ag^+(aq) + e^- \rightarrow Ag(s)$ $E^\circ = +0.80$ V

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$Fe^{2+}(aq) + 2e^- \rightarrow Fe(s)$ $E^\circ = -0.44$ V

What is the emf of the given reaction?

$Fe(s) + 2Ag^+(aq) \rightarrow Fe^{2+}(aq) + 2Ag(s)$

(a) 1.16 V
(b) 1.24 V
(c) 2.04 V
(d) -1.16 V

9. Which of the following metal is resistant to corrosion?
   (a) Titanium
   (b) Zinc
   (c) Magnesium
   (d) Iron

10. In electrochemical corrosion
   (a) Anode undergoes oxidation
   (b) Cathode undergoes oxidation
   (c) Both undergo oxidation
   (d) None undergoes oxidation

Part – B (2 x 10 = 20 Marks)

11. (a) i. What is electron affinity? What are the factors influencing it and arrange Br, F, I and Cl in the order of increasing electron affinity. (4 Marks)

ii. According to Pearson's rule distinguish between (6 Marks)
A) Hard acid and soft acid B) Hard base and soft base

(OR)

(b) i. Calculate the Gibbs free energy for a reaction which involves the change in enthalpy as 19.07 Kcal and change in entropy as 90 cal/K at 293K (4 Marks)

ii Derive Nernst equation and give its applications. (6 Marks)

12. (a) i. Give an account of Hydrogen evolution type of wet corrosion (4 Marks)

ii. What is the expression for the solubility product constant of common salt? What factors affect the value of Ksp? (6 Marks)

(OR)

(b) i. What is Internal energy? Give its relation with enthalpy.

ii. What is the Cell Potential of the electrochemical cell for the given cell reaction?
$Pb^{2+} + Cd \rightarrow Pb + Cd^{2+}$
Given that E^0 cell = 0.277 volts, Temperature = 25°C, $[Cd^{2+}] = 0.02$ M, and $[Pb^{2+}] = 0.2$ M

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Set- 1

INTERNAL ASSESSMENT – II [FJ-II]

Program	: B. Tech.	Date	: 21-11-2024
Course Code & Title	: 21CYB101J & Chemistry	Time	: 8.00-9.00am
Year & Sem.	: I Year & I Sem.	Max. Marks	: 30 marks

Course Articulation Matrix

At the end of this course, learners will be able to:	POs				
Course Outcomes (CO)	1	2	3	4	5
CO1 Rationalize bulk properties using periodic properties of elements, evaluate water quality parameters like hardness and alkalinity	3		3	2	
CO2 Utilize the concepts of thermodynamics in understanding thermodynamically driven chemical reactions, determine acidic strength and redox potentials of aqueous solution	3	3	3		
CO3 Perceive the importance of stereochemistry in synthesizing organic molecules applied in pharmaceutical industries, determine acidic strength and conductance of aqueous solution		3	3	2	
CO4 Utilize the concepts of polymer processing for various technological applications, determine average molecular weight of the polymer	3		3	3	
CO5 Analyze the importance of advanced processing techniques towards engineering applications and measure the acidic strength of aqueous solution	3		3		3

Part-A (10X1 = 10 Marks)
Answer ALL the questions

1. When equilibrium is reached inside the two half-cells of the electrochemical cell, what is the net voltage across the electrodes?
   (a) > 1 (b) < 1 (c) = 0 (d) Not defined
2. What will be the Gibbs free energy for the reaction of conversion of ATP into ADP at 293 Kelvin, if the change in enthalpy is 19.07 Kcal and the change in entropy is 90 cal per Kelvin.
   (a) 7.3 cal (b) -5.3 cal (c) 7.3 Kcal (d) -7.3 Kcal
3. An oxide layer is considered as protective if ratio of volume of oxide layer to volume of base metal is
   (a) > 1 (b) > 10 (c) < 1 (d) > 3
4. Which of the following factors does not affect the electrode potential of an electrode?
   (a) Nature of the electrode (metal) (b) Temperature of the solution
   (c) Molarity of the solution (d) Size of the electrode
5. Which of the following statement is incorrect?
   (a) At constant pressure, $\Delta H = \Delta E + P\Delta V$
   (b) Thermodynamic symbol for entropy is S.
   (c) Gibbs free energy is a state function.
   (d) For an endothermic process, ΔH is negative.
6. Rusting of iron in acidic aqueous solution of electrolyte occurs with the evolution of _____ gas.
   (a) Nitrogen (b) Chlorine (c) Oxygen (d) Hydrogen

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7. According to the convention, the Daniel cell is represented as:
(a) Zn | $ZnSO_4$ || $CuSO_4$ | Cu, E = 1.09 volt
(b) Zn | $ZnSO_4$ || Cu | $CuSO_4$, E = 1.09 volt
(c) $ZnSO_4$ | Zn || $CuSO_4$ | Cu, E = 1.09 volt
(d) Zn | ZnS || $CuSO_4$ | Cu, E = 1.09 volt

8. What type of reaction takes place upon treatment of a ketone with HCN to form a cyanohydrin?
(a) Nucleophilic addition (b) Nucleophilic substitution
(c) Electrophilic addition (d) Electrophilic substitution

9. How many aromatic isomers of dibromobenzene exist?
(a) 2 (b) 3 (c) 4 (d) 6

10. Which of the following is true about Fischer Projection?

(a) The vertical lines are oriented away from you and the horizontal lines are oriented toward you.
(b) The vertical lines are oriented towards you and the horizontal lines are oriented away from you.
(c) Both the horizontal and vertical lines are oriented away from you.
(d) Both the horizontal and vertical lines are oriented towards you.

Part-B (2 x 10 = 20 Marks)

11(a) (i) Derive the expression for the Helmholtz free energy and explain its physical significance. (6 Marks)

(ii) For a Daniel Cell involving a cell reaction,

$$Zn_{(s)} + Cu^{2+}_{(aq)} \leftrightarrow Zn^{2+}_{(aq)} + Cu_{(s)}$$

The standard free energy of formation of $Zn_{(s)}$, $Cu_{(s)}$, $Cu^{2+}_{(aq)}$, $Zn^{2+}_{(aq)}$ are 0, 0, 64.4kJ/mol and -154 kJ/mol respectively. Calculate the standard EMF of the cell. (4 Marks)

(OR)

11(b) (i) Describe potentiometric redox titration with an example and explain its advantages. (6 Marks)

(ii) Differentiate between dry corrosion and wet corrosion. (4 Marks)

12(a) Write a note on the Pourbaix diagram of iron (Fe). (10 Marks)

(OR)

12(b) (i) Explain the SN^1 mechanism with a suitable example. (6 Marks)

(ii) Draw the Newman and Sawhorse projections for the eclipsed and staggered conformations of ethane. Which conformation is more stable and why? (4 Marks)

No.	BL	CO	PO
1	2	2	2
2	4		2
3	1		1
4	2		2
5	1		3
6	4		2
7	2		1
8	3	3	2
9	2		3
10	3		4

Q. No.	BL	CO	PO
11a. (i)	1	2	1
(ii)	2		2
11b. (i)	3		3
(ii)	1		1
12a.	2		2
12b. (i)	3	3	2
(ii)	2		3

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Set- 2

INTERNAL ASSESSMENT – II [FJ-II]

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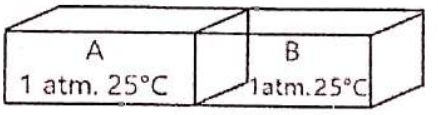
Program	: B. Tech.	**Date**	: 21-11-2024
Course Code & Title	: 21CYB101J & Chemistry	**Time**	: 8.00-9.00am
Year & Sem.	: I Year & I Sem.	**Max. Marks**	: 30 marks

Course Articulation Matrix

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CO3 Perceive the importance of stereochemistry in synthesizing organic molecules applied in pharmaceutical industries, determine acidic strength and conductance of aqueous solution		3	3	2	
CO4 Utilize the concepts of polymer processing for various technological applications, determine average molecular weight of the polymer	3		3	3	
CO5 Analyze the importance of advanced processing techniques towards engineering applications and measure the acidic strength of aqueous solution	3		3		3

Part-A (10X1 = 10Marks)
Answer ALL the questions

1. The equilibrium constant for a cell reaction, $Cu_{(g)} + 2Ag^{+}_{(aq)} \leftrightarrow Cu^{2+}_{(aq)} + 2Ag_{(s)}$ is 4×10^{16}. Find $E^{\circ}_{(cell)}$ for the cell reaction.
   (a) 0.63 V (b) 0.49 V (c) 1.23 V (d) 3.24 V
2. An isolated box, equally partitioned, contains two ideal gasses A and B as shown. When the partition is removed, the gases mix. The changes in enthalpy (ΔH) and entropy (ΔS) in the process respectively are:



   (a) Zero, positive (b) Zero, negative (c) Positive, zero (d) Negative, zero
3. Passivity is not reason for inertness of the following:
   (a) Au (b) Al (c) Ti (d) Ni
4. In a reaction, the change in entropy is given as 2.4 cal/K and the change in Gibbs free energy is given as 3.4 kcal, then the change in heat at the temperature of 20 °C is:
   (a) 4.1 cal (b) 3.4 cal (c) 3.4 Kcal (d) 4.1 Kcal
5. Mercury covered by a layer of mercurous chloride in contact with saturated potassium chloride solution is a description of which of the following types of electrodes?
   (a) Chlorine (b) Potassium (c) Calomel (d) Silver/Silver chloride

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6. A process is carried out at constant volume and at constant entropy. It will be spontaneous if:
(a) $\Delta H < 0$ (b) $\Delta U < 0$ (c) $\Delta A < 0$ (d) $\Delta G < 0$

7. The entropy of an isolated system always _____ and reaches _____ when equilibrium is reached.
(a) remains constant, maximum (b) decreases, minimum
(c) increases, maximum (d) decreases, constant

8. The isomers which can be inter converted through rotation around a single bond are termed as:
(a) Tautomers (b) positional isomers (c) conformers (d) diastereomers

9. Which of the following statements is **not** correct?
(a) Fischer projection represents the molecule in an eclipsed conformation
(b) Newman projection can be represented in eclipsed, staggered and skew conformations
(c) Fischer projection of the molecule is its most stable conformation
(d) In Sawhorse projections, the lines are inclined at an angle of 120° to each other

10. The reactivity order of alkyl halides in SN^1 mechanism is
(a) $CH_3X > 1^o > 2^o > 3^o$ (b) $CH_3X < 1^o < 2^o < 3^o$
(c) $CH_3X > 3^o > 2^o > 1^o$ (d) $CH_3X > 3^o > 1^o > 2^o$

Part-B (2 x 10 = 20 Marks)

11(a) (i) Describe the construction and working of an electrochemical cell, provide an example to illustrate its operation. (6 Marks)

(ii) Calculate the entropy change for methane combustion from the given thermodynamic data. The Standard entropies of the substances involved in the above reaction are: $\Delta S°$(J/K. mol)) of $CH_{4(g)}$ =186, $O_{2\,(g)}$ = 205 $CO_{2\,(g)}$ = 214 and $H_2O_{(l)}$ = 70 (4 Marks)

(OR)

11(b) (i) Derive Nernst equation and explain its significance in electrochemistry. (6 Marks)

(ii) Define entropy and explain its significance in thermodynamic and natural processes. (4Marks)

12(a) Explain mechanisms of dry corrosion and wet corrosion, providing the chemical reaction involved in each process. (10 Marks)

(OR)

12(b) (i) Explain the addition of HBr to propylene, in the presence and absence of peroxide. (8 Marks)

(ii) Define diastereomers with an example. (2 Marks)

Q. No.	BL	CO	PO
1	2	2	2
2	4		2
3	1		1
4	2		2
5	1		3
6	4		2
7	2		1
8	3	3	2
9	2		3
10	3		4

Q. No.	BL	CO	PO
11a (i)	1	2	1
(ii)	2		2
11b (i)	3		3
(ii)	1		1
12a.	2		2
12b (i)	3	3	3
(ii)	2		4

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SRM

DEPARTMENT OF CHEMISTRY
College of Engineering and Technology
SRM Institute of Science and Technology
Kattankulathur – 603203

Set- 2

INTERNAL ASSESSMENT – II [FJ2]

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & I Sem
Course Articulation Matrix

Date: 25/11/2024
Time: 12.30 - 1.30 pm.
Max. Marks: 30 Marks

At the end of this course, learners will be able to:	POs				
Course Outcomes (CO)	1	2	3	4	5
CO1 Rationalize bulk properties using periodic properties of elements, evaluate water quality parameters like hardness and alkalinity	3		3	2	
CO2 Utilize the concepts of thermodynamics in understanding thermodynamically driven chemical reactions, determine acidic strength and redox potentials of aqueous solution	3	3	3		
CO3 Perceive the importance of stereochemistry in synthesizing organic molecules applied in pharmaceutical industries, determine acidic strength and conductance of aqueous solution		3	3	2	
CO4 Utilize the concepts of polymer processing for various technological applications, determine average molecular weight of the polymer	3		3	3	
CO5 Analyze the importance of advanced processing techniques towards engineering applications and measure the acidic strength of aqueous solution	3		3		3

Part – A (10 x 1 = 10 Marks)
Answer ALL the Questions

1 In a reversible process, entropy of the system
a) increases b) decreases
c) zero d) remains constant

2 The name of the equation showing relation between electrode potential standard potential (E°) and concentration of ions in solution is
a) Kohlrausch equation b) Nernst equation
c) Faraday's equation d) Ohm's equation

3 Which of the following statement is incorrect?
a) At constant pressure, $\Delta H = \Delta E + P\Delta V$ b) thermodynamic symbol for entropy is S.
c) Gibbs free energy is a state function. d) For an endothermic process, ΔH is negative.

4 All of the following have a standard heat of formation value of zero at 25 °C and 1.0 atmosphere except:
a) $N_2(g)$ b) Fe(s)
c) Ne(g) d) H(g)

5 Passivity is due to
a) Higher EMF b) Lower EMF
c) Oxide film d) Carbon film

6 In Pourbaix diagrarn the redox reaction, $Fe^{2+}+2e^- \rightarrow Fe_{(s)}$ is
a) pH dependent b) pH independent
c) solvent dependent d) solvent independent

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7 For the reduction of silver ions with copper metal the standard cell potential was found to be +0.46 V at 25° C. The value of standard Gibbs energy, ΔG° will be ($F = 96500\ C\ mol^{-1}$) ------.

a) -44.5 kJ b) -98.0 kJ

c) -89.0 kJ d) -89.0 J

8 According to the Cahn Ingold Prelog selection rules, the decreasing order of preference is

a) $-CH(CH_3)_2 > -C_6H_5 > -H > -NH_2$

b) $-NH_2 > -C_6H_5 > -CH(CH_3)_2 > -H$

c) $-NH_2 > -CH(CH_3)_2 > -C_6H_5 > -H$

d) $-C_6H_5 > -CH(CH_3)_2 > -NH_2 > -H$

9 How many aromatic isomers of dichlorobenzene exist?

a) 2 b) 3

c) 4 d) 6

10 When an organic molecule has a plane of symmetry, it will be optically

a) inactive b) active

c) step-by-step optically active d) partially active

Part – B (2 x 10 = 20 Marks)

11. a. Derive Gibbs-Helmholtz equation and give its importance. (10 Marks)

(OR)

b. Explain the mechanism involved in dry and chemical corrosion. (10 Marks)

12 a. i Calculate the entropy change for methane combustion. (5 Marks)
From the Thermodynamic Data, the Standard entropies of the substances involved in the above reaction are: ΔS°(J/K. mol)) of $CH_4(g) = 186$, $O_2(g) = 205$ $CO_2(g) = 214$ and $H_2O(l) = 70$

ii. Write a short note on Galvanic cell with diagram (5 Marks)

(OR)

b. i. Discuss any two types of structural isomers with examples. (6 Marks)

ii. Draw the conformational analysis of n-butane with potential energy plot. (4 Marks)

Q.No	BL	CO	PO
1	1	2	1
2	2		2
3	2		2
4	2		3
5	1		3
6	3		2
7	3		2
8	4	3	2
9	3		2
10	2		1

Q.No	BL	CO	PO
11a.	4	2	3
11b.	3		3
12a. i.	3		2
ii.	2		3
12b.i.	2	3	2
ii.	1		3

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SRM

DEPARTMENT OF CHEMISTRY
College of Engineering and Technology
SRM Institute of Science and Technology
Kattankulathur – 603203

Set- 1

INTERNAL ASSESSMENT – II [FJ2]

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & I Sem

Date: 25/11/2024
Time: 12.30 - 1.30pm.
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Course Articulation Matrix

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CO4 Utilize the concepts of polymer processing for various technological applications, determine average molecular weight of the polymer	3		3	3	
CO5 Analyze the importance of advanced processing techniques towards engineering applications and measure the acidic strength of aqueous solution	3		3		3

Part – A (10 x 1 = 10 Marks)
Answer ALL the Questions

1 For a reversible process $\Delta_{sys} + \Delta_{surr}$ is
a) > 0 b) < 0
c) ≥ 0 d) = 0

2 The following are state functions except
a) enthalpy b) heat
c) internal energy d) entropy

3 Which molecule has zero standard molar enthalpy of formation at 298 K
a) $Cl_2(g)$ b) H_2O
c) Br_2 (g) d) CH_4 (g)

4 If the enthalpy change for a reaction is zero, ΔG° is equal to
a) $T\Delta S^\circ$ b) $-T\Delta S^\circ$
c) $-\Delta H^\circ$ d) ln Keq

5 A process is carried out at constant volume and at constant entropy. It will be spontaneous if:
a) $\Delta H < 0$ b) $\Delta U < 0$
c) $\Delta H = 0$ d) $\Delta G < 0$

6 The standard electrode potentials (E°) for Fe^{3+}/Fe^{2+} and Fe^{2+}/Fe electrodes are + 0.77 V and - 0.44 V respectively at 300 K. The E° of Fe^{3+}/Fe electrode at the same temperature is
a) – 0.11 V b) 1.21 V
c) 0.036 V d) – 0.036 V

7 Anhydrous inorganic liquid metal surface in absence of moisture undergoes --------

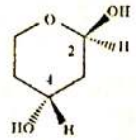
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a) Wet corrosion b) Dry corrosion
b) Galvanic corrosion d) Pitting corrosion

8 The conformations n-butane commonly known as gauche, eclipsed and anti-conformations can be inter-converted by rotation around
a) C-H bond of methyl group b) C1-C2 linkage
c) C2-C3 linkage d) C-H bond of methylene group

9 Arrange the following in the decreasing order of leaving group in nucleophilic substitution reaction.
a) $H^- > Cl^- > HO^- > Br^- > CH_3COO^-$ b) $Cl^- > Br^- > HO^- > H^- > CH_3COO^-$
c) $Cl^- > Br^- > CH_3COO^- > HO^- > H^-$ d) $HO^- > CH_3COO^- > H^- > Br^- > Cl^-$

10 Which is the correct assignment of chirality at C2 and C4 of the following molecule?



a) 2S,4S b) 2R,4R
c) 2S,4R d) 2R,4S

Part – B (2 x 10 = 20 Marks)

11. a. Derive Nernst equation and give its application in acid-base and redox reactions. (10 Marks)

(OR)

b. i. Calculate the enthalpy change for methane combustion. (5 Marks)
ΔH°_f values for $CH_4(g) = 75$, $O_2(g) = 0$, $H_2O(l) = 286$, $CO_2(g) = 394$ KJ mol^{-1} respectively

ii. Define enthalpy and internal energy and its relation. (5 Marks)

12 a. With a neat sketch explain Pourbaix diagram for Iron. (10 Marks)

(OR)

b. i. Write a note on optical isomerism by taking suitable example. (5 Marks)

ii. Explain S_N1 mechanism of nucleophilic substitution reaction. (5 Marks)

Q.No	BL	CO	PO
1	2	2	2
2	1		1
3	1		2
4	2		1
5	3		2
6	4		3
7	2		3
8	2	3	4
9	4		3
10	3		2

Q.No	BL	CO	PO
11a.	3	2	3
11b.i.	3		2
ii.	2		3
12a.	4		4
12b.i.	3	3	2
ii.	1		3

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